PHY222 Classwork

StdentNo: 4503127

Question 1:

1. Solve the differential equation analytically to find ( ).

$$\frac{dN(t)}{dt} = -\lambda N(t)$$

$$\frac{dN(t)}{N(t)} = -\lambda\, dt$$

$$\int \frac{dN(t)}{N(t)} = -\int \lambda\, dt$$

$$\ln|N(t)| = -\lambda t + C$$

$$N(t) = e^{-\lambda t + C}$$

$$N(t) = e^{-\lambda t} e^{C}$$

$$e^{C} = A$$

$$N(t) = Ae^{-\lambda t}$$

$$N(0) = 100, \qquad \lambda = 0.1$$

$$100 = Ae^{0}$$

$$A = 100$$

$$N(t) = 100e^{-0.1t}$$

2. Plot ( ) over a period of 50 days using Python in a Jupyter notebook.

Imports

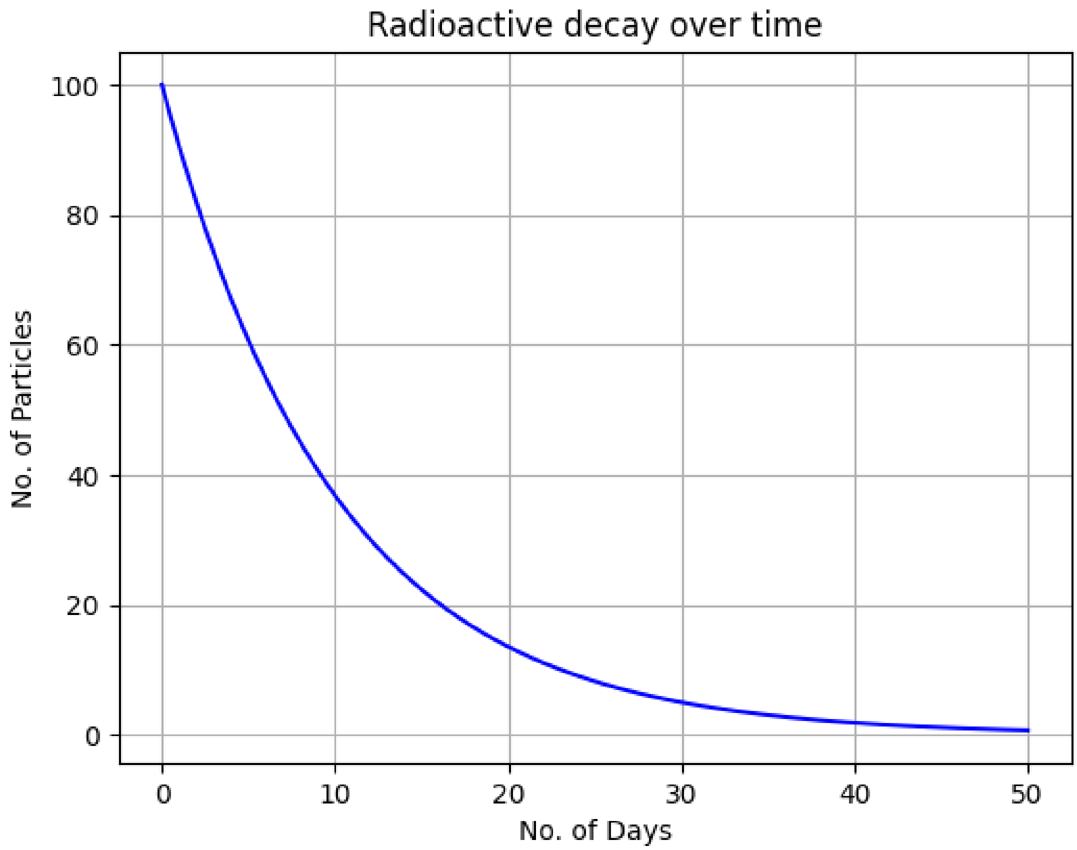
```
import numpy as np
import matplotlib.pyplot as plt

def N(t):
    A=100
    lam=0.1
    return A* np.exp(-lam*t)

t = np.linspace(0, 50, 1000)

y =N(t)

plt.plot(t,y, color='blue')
plt.title('Radioactive decay over time')
plt.grid(True)
plt.xlabel('No. of Days')
plt.ylabel('No. of Particles')
plt.show()
```



Question 2

1. Solve the differential equation analytically to find ( ).

$$
\begin{aligned}
\frac{dT(t)}{dt} &= -k(T(t) - T_{amb}) \\
\frac{dT(t)}{(T(t) - T_{amb})} &= -kdt \\
\int \frac{dT(t)}{T(t)} &= \int -kdt \\
\ln |T(t) - T_{amb}| &= -kt + C \\
T(t) &= e^{-kt+C} + T_{amb} \\
T(t) &= Ae^{-kt} + T_{amb}
\end{aligned}
$$

So, $T(0) = 80^{\circ}\mathrm{C}$ $T_{amb} = 25^{\circ}\mathrm{C}$ \$ k = 0.1 min^{-1}\$

$$\begin{aligned}
T(0) &= Ae^{-k(0)} + T_{amb} \\
80^{\circ}\mathrm{C} &= Ae^{-k(0)} + 25^{\circ}\mathrm{C} \\
A &= 55
\end{aligned}$$

$$T(t) = 55^{\circ}\mathrm{C}\, e^{-0.1t} + 25^{\circ}\mathrm{C} \quad \text{where } t \text{ is in minutes}$$

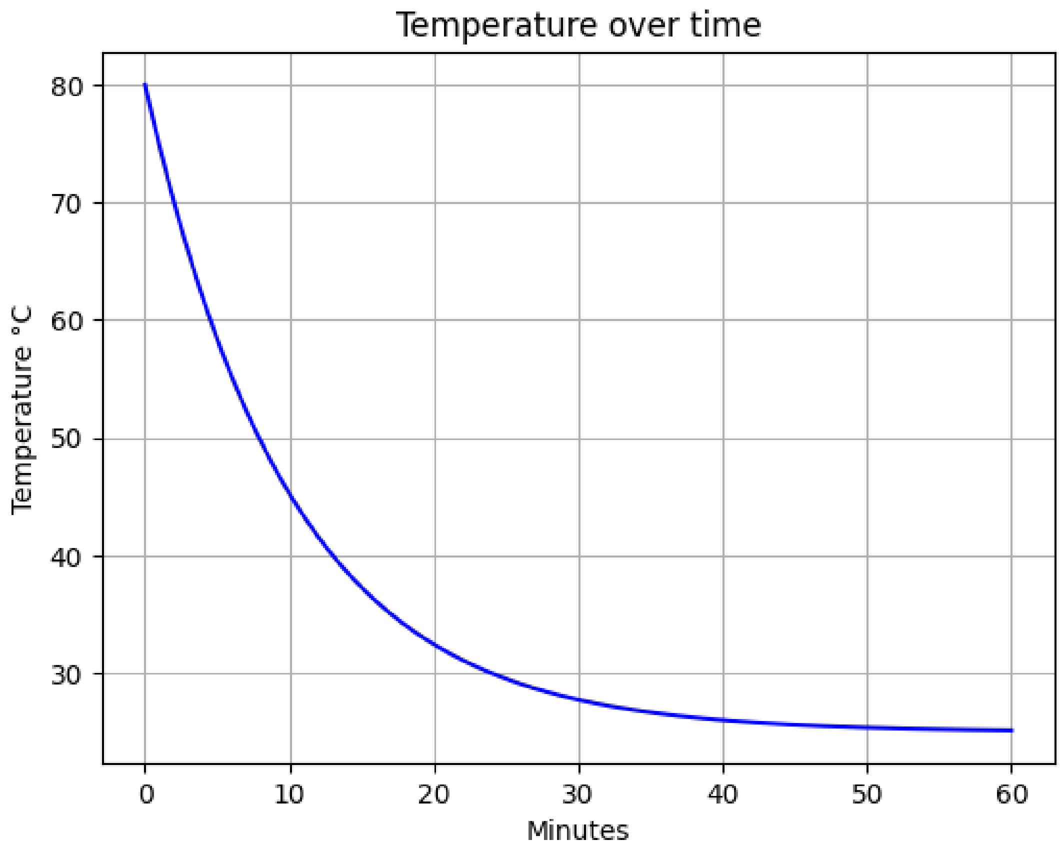
2. Plot ( ) over a period of 60 minutes using Python in a Jupyter notebook.

```
def T(t):
    return 55*np.exp(-0.1*t) + 25

t = np.linspace(0, 60, 1000)

y =T(t)

plt.plot(t,y, color='blue')
plt.title('Temperature over time')
plt.grid(True)
plt.xlabel('Minutes')
plt.ylabel('Temperature \u00B0C')
plt.show()
```


Temperature over time
80
70
60
50
40
30
Temperature °C
0
10
20
30
40
50
60
Minutes